Cambridge
O Level

Cambridge International Examinations
Cambridge Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

6421227531

MATHEMATICS (SYLLABUS D) **4024/11**

Paper 1 **May/June 2016**

2 hours

Candidates answer on the Question Paper.

Additional Materials: Geometrical instruments

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

If working is needed for any question it must be shown in the space below that question.
Omission of essential working will result in loss of marks.

ELECTRONIC CALCULATORS MUST NOT BE USED IN THIS PAPER.

The number of marks is given in brackets [ ] at the end of each question or part question.
The total of the marks for this paper is 80.

This document consists of **20** printed pages.

© UCLES 2016

CAMBRIDGE International Examinations

ELECTRONIC CALCULATORS MUST NOT BE USED IN THIS PAPER.

1 **(a)** Evaluate $12 - 6 \div 3 + 4$.

Answer ..... [1]

(b) Evaluate 0.3×1.5.

Answer ..... [1]

---

2 **(a)** Evaluate $\frac{2}{3} - \frac{5}{8}$.

Answer ..... [1]

(b) Evaluate $\frac{1}{3} \div \frac{7}{9}$, giving your answer as a fraction in its lowest terms.

Answer ..... [1]

© UCLES 2016

3 **(a)** An aircraft leaves at 22 35 on a flight that takes 3 hours and 50 minutes.

Find the time when the aircraft arrives.

Answer ..... [1]

(b) The aircraft flies a distance of 3200 km, correct to the nearest 100 km.

Write down the lower bound for the distance.

Answer ..... km [1]

---

4 A bottle full of liquid has a total mass of 1.27 kg.
When the bottle is half-full of liquid the total mass is 900 grams.

Calculate the mass of the bottle.

Answer ..... grams [2]

---

5 Stella walks to a park.

For 4 minutes she walks at a rate of 80 steps per minute.
For 1 minute she walks at a rate of 120 steps per minute.

Find the mean number of steps per minute she takes.

Answer ..... [2]

---

6 **(a)** Write the number 0.034×10^{-3} in standard form.

Answer ..... [1]

(b) Arrange the following numbers in order, starting with the smallest.

0.034×10^{-3} 33.7×10^{-6} 0.42×10^{-5}

Answer ..... , ... , ... [1]
smallest

© UCLES 2016

7 By writing each number correct to 1 significant figure, estimate the value of

$$\frac{29.2 \times 8.17}{0.396}.$$

Answer ..... [2]

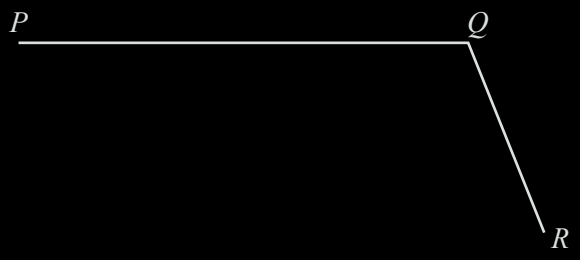
---

8 **(a)** Complete the diagram to make a quadrilateral $ABCD$ which has AC as its line of symmetry.



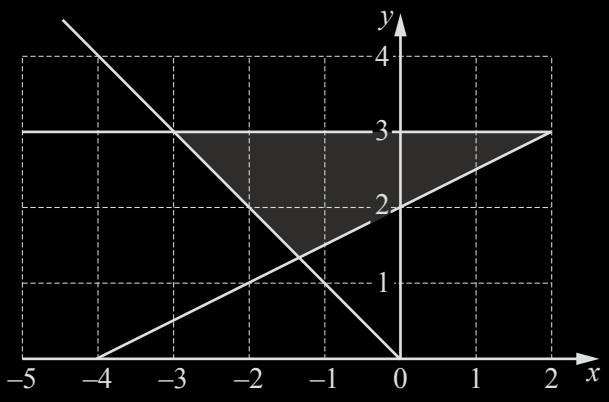
[1]

(b) Complete the diagram to make a quadrilateral $PQRS$ which has rotational symmetry of order 2.



[1]

9



The shaded region in the diagram is defined by three inequalities.

One of these is $y \geqslant \frac{1}{2}x + 2$.

Write down the other two inequalities.

Answer .....

..... [2]

---

10 Factorise completely $3xy - 20 + 5x - 12y$.

Answer ..... [2]

© UCLES 2016

11 $f(x) = 2x - 9$

(a) Find $f\left(-\frac{3}{4}\right)$.

Answer ..... [1]

(b) Find $f^{-1}(3)$.

Answer ..... [2]

---

12 A map is drawn to a scale of 2 cm to 5 km.

(a) Two towers are 9 km apart.

Calculate the distance between them on the map.

Answer ..... cm [1]

(b) On the map, a town covers an area of $4\,cm^2$.

Calculate its actual area.

Answer ..... km^2 [1]

(c) Express the scale of the map in the form 1 : n.

Answer 1 : ..... [1]

---

13 Solve the simultaneous equations.

$$3x = 4y$$
$$1 + 5x = 6y$$

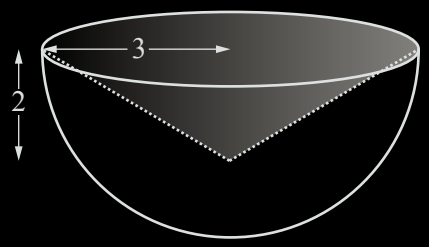
Answer $x =$ .....

$y =$ ..... [3]

© UCLES 2016

14 **[The volume of a sphere is $\frac{4}{3}\pi r^3$] [The volume of a cone is $\frac{1}{3}\pi r^2 h$]**

A cone is removed from a solid wooden hemisphere of radius 3 cm.



The cone has radius 3 cm and height 2 cm.

The volume of wood remaining is $k\pi$ cm^3.

Find k.

Answer $k =$ ..... [3]

---

15 **(a)** y is directly proportional to the square of x.

Given that $y = 8$ when $x = 4$, find y when $x = 3$.

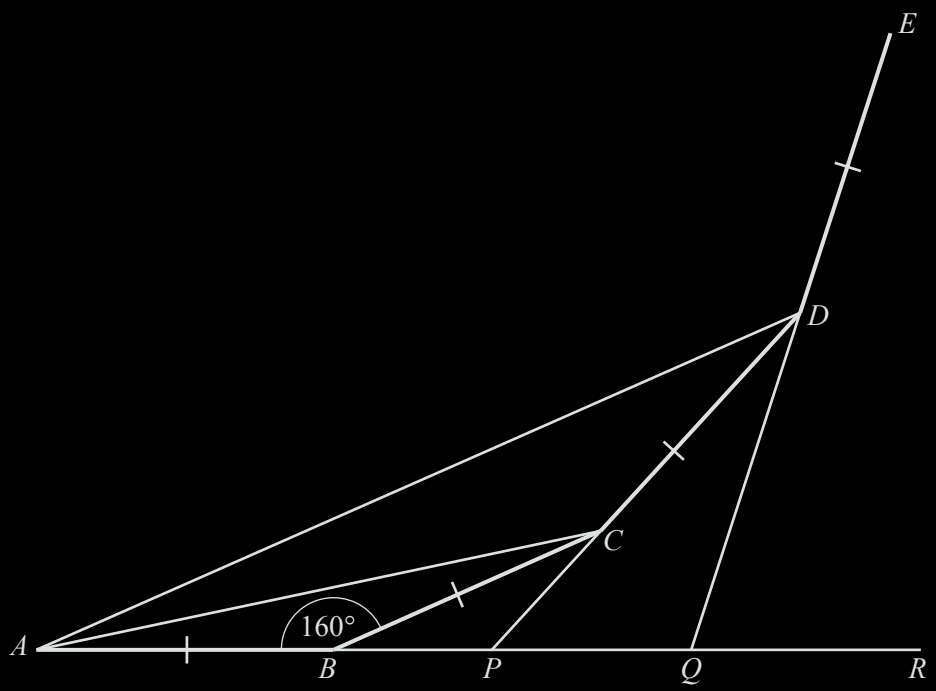
Answer $y =$ ..... [2]

(b) p is inversely proportional to q.
It is known that $p = 15$ for a particular value of q.

Write down the value of p when this value of q is doubled.

Answer $p =$ ..... [1]

16



In the diagram, AB, BC, CD and DE are four sides of a regular polygon.
Each interior angle of the polygon is 160°.

$ABPQR$, DCP and EDQ are straight lines.

(a) Find $C\hat{A}B$.

Answer $C\hat{A}B =$ ..... [1]

(b) Find $C\hat{B}P$.

Answer $C\hat{B}P =$ ..... [1]

(c) Find $D\hat{Q}R$.

Answer $D\hat{Q}R =$ ..... [1]

© UCLES 2016

17 A sequence of diagrams is made using counters.

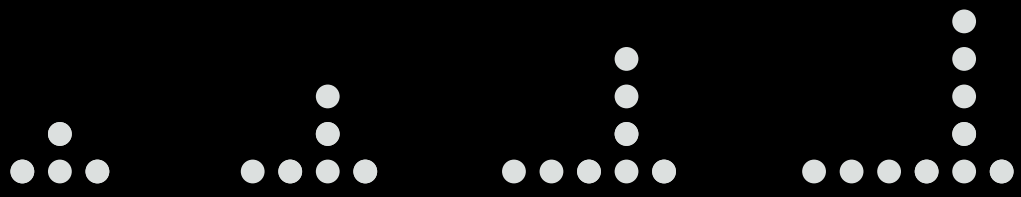


Diagram 1 Diagram 2 Diagram 3 Diagram 4

(a) Complete the table.

Diagram number	1	2	3	4	5
Number of counters	4	6	8		

[1]

(b) Find an expression, in terms of n, for the number of counters in Diagram n.

Answer ..... [1]

(c) In this sequence, Diagram p has 200 counters.

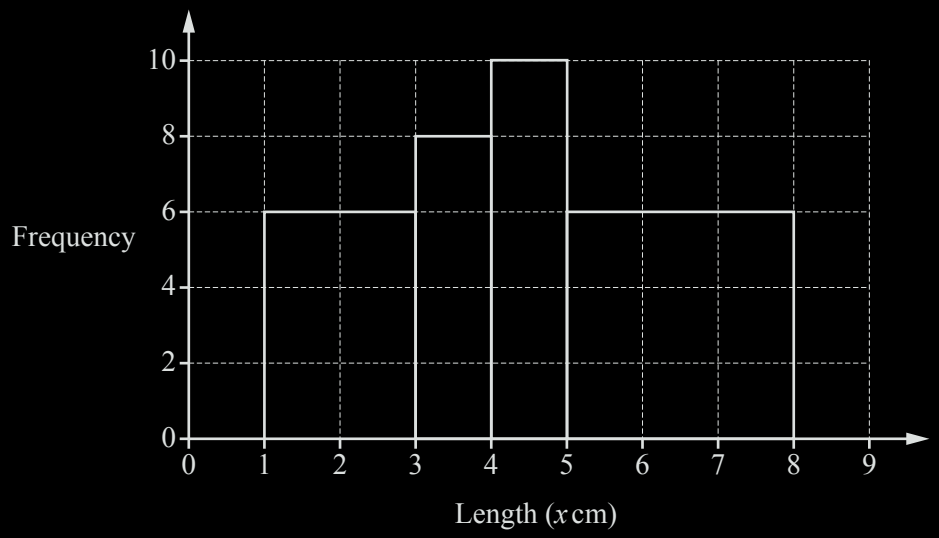
Find the value of p.

Answer $p =$ ..... [2]

18 Henri did a survey of the lengths of the leaves on a plant.
The results are summarised in the table.

Length (x cm)	$1 < x \leqslant 3$	$3 < x \leqslant 4$	$4 < x \leqslant 5$	$5 < x \leqslant 8$
Frequency	6	8	10	6

(a) When asked to draw a histogram to illustrate the results, Henri drew the following diagram.

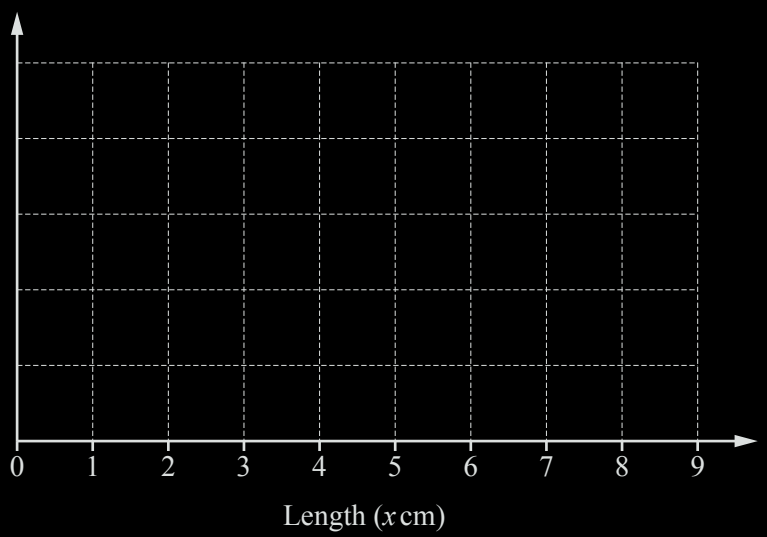


Explain why this diagram is incorrect.

.....

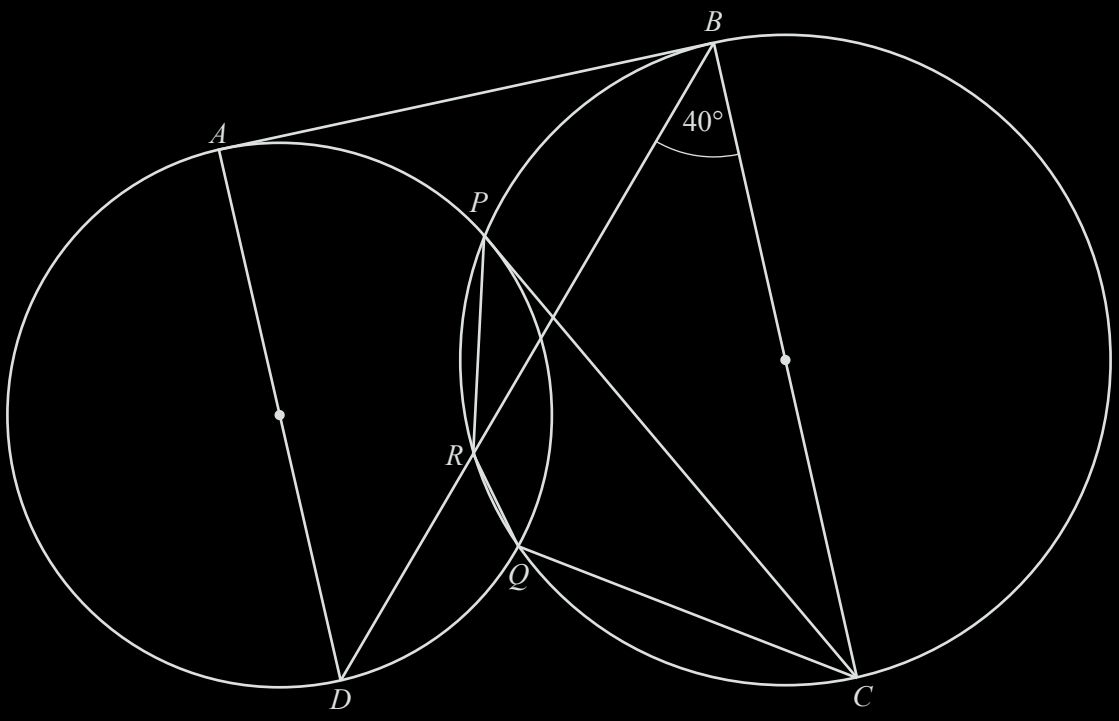
..... [1]

(b) On the grid below, draw a correct histogram for Henri's results.



[3]

19



In the diagram, the two circles intersect at P and Q.
The line AB is a tangent to the circles at A and B.
AD and BC are diameters.
BD intersects the larger circle at R.

$D\hat{B}C = 40°$.

(a) Find $C\hat{P}R$.

Answer $C\hat{P}R =$ ..... [1]

(b) Find $C\hat{Q}R$.

Answer $C\hat{Q}R =$ ..... [1]

(c) Find $A\hat{B}D$.

Answer $A\hat{B}D =$ ..... [1]

(d) Find $A\hat{D}B$.

Answer $A\hat{D}B =$ ..... [1]

© UCLES 2016

20 The number of goals scored in each of 50 football matches was recorded.
The results are given in the table.

Number of goals scored	0	1	2	3	4	5	6
Frequency	16	11	9	7	6	0	1

For these results, find

(a) the mode,

Answer ..... [1]

(b) the median,

Answer ..... [1]

(c) the mean.

Answer ..... [2]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge International Examinations Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cie.org.uk after the live examination series.

Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.

© UCLES 2016

21 **(a)** Express 500 as the product of its prime factors.

Answer ..... [1]

(b) $M = 2 \times 3^2$ $N = 2^4 \times 3^2$

Find the values of p and q when

(i) $M \times N = 2^p \times 3^q$,

Answer $p =$ ..... $q =$ ..... [1]

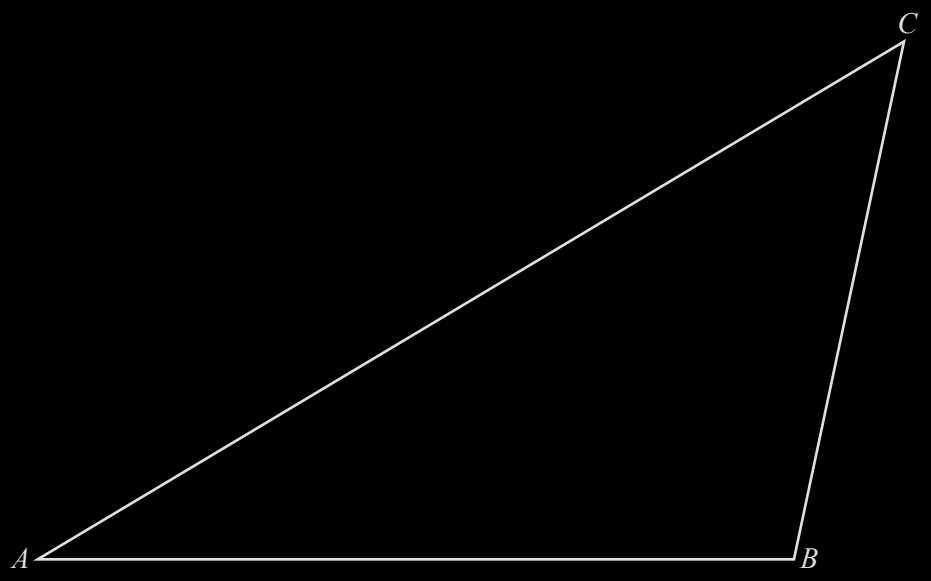
(ii) $M \div N = 2^p \times 3^q$,

Answer $p =$ ..... $q =$ ..... [1]

(iii) $N^2 = 2^p \times 3^q$.

Answer $p =$ ..... $q =$ ..... [1]

22 The diagram shows triangle ABC.



(a) Measure $A\hat{B}C$.

Answer $A\hat{B}C$ = ... [1]

(b) On the diagram, construct the locus of points, **inside** triangle ABC, that are

(i) 4 cm from B, [1]

(ii) 2 cm from AC. [1]

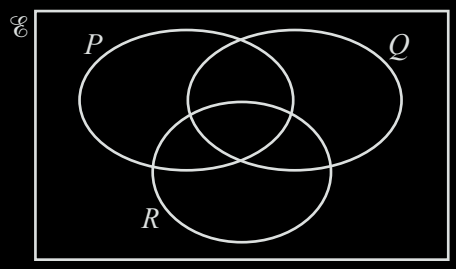
(c) The point P is
4 cm from B,
2 cm from AC,
and nearer to A than to C.

Label the position of P on the diagram and find the length of AP.

Answer AP = ... cm [1]

© UCLES 2016

23 **(a)** In the Venn diagram, shade the region which represents the subset $(P \cup Q)' \cap R$.



[1]

(b) $\mathscr{E} = \{ x : x \text{ is an integer and } 22 \leqslant x \leqslant 33 \}$
$E = \{ x : x \text{ is an even number} \}$
$T = \{ x : x \text{ is a multiple of 3} \}$
$F = \{ x : x \text{ is a multiple of 4} \}$

(i) List the members of $T \cap F$.

Answer ..... [1]

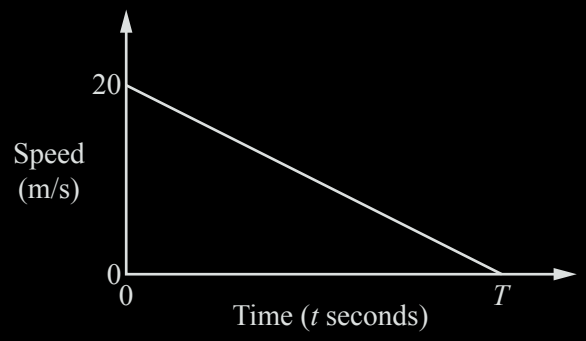
(ii) Find $\mathrm{n}(E \cup T)$.

Answer ..... [1]

(iii) Given that $y \in F' \cap E$, find one possible value of y.

Answer y = ... [1]

24 The diagram shows the speed-time graph of a train which slows down from 20 m/s to a stop in T seconds.



(a) (i) Find an expression, in terms of T, for the retardation of the train.

Answer ..... m/s^2 [1]

(ii) Find the speed of the train when $t = \frac{3}{4}T$.

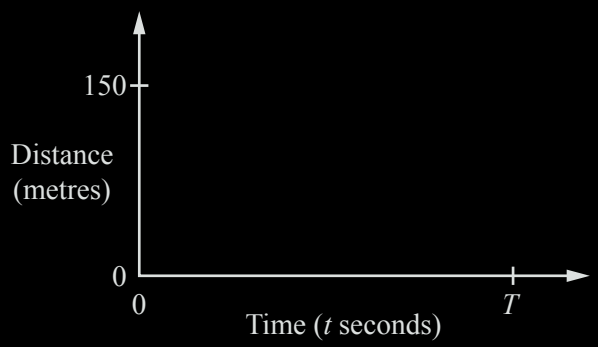
Answer ..... m/s [1]

(b) The distance travelled by the train between $t = 0$ and $t = T$ is 150 m.

(i) Find T.

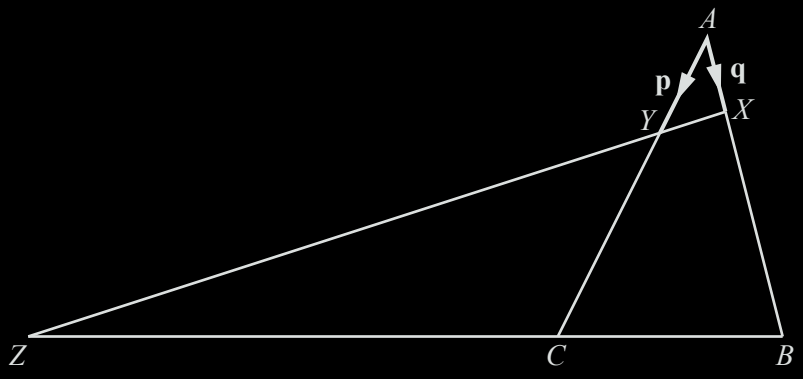
Answer T = ... [1]

(ii) On the diagram, sketch the distance–time graph of the train.



[1]

25



In the diagram,

X is the point on AB where $AX = \frac{1}{4}AB$,

Y is the point on AC where $AY = \frac{1}{3}AC$,

Z is the point on BC produced where $CZ = 2BC$.

$\overrightarrow{AY} = \mathbf{p}$ and $\overrightarrow{AX} = \mathbf{q}$.

(a) Express, as simply as possible, in terms of **p** and **q**,

(i) $\overrightarrow{XY}$,

Answer $\overrightarrow{XY}$ = ... [1]

(ii) $\overrightarrow{BC}$,

Answer $\overrightarrow{BC}$ = ... [1]

(iii) $\overrightarrow{XZ}$.

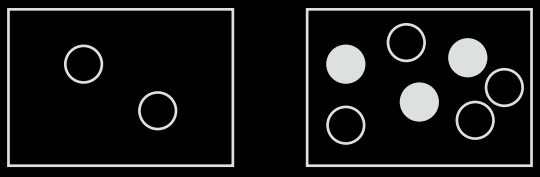
Answer $\overrightarrow{XZ}$ = ... [2]

(b) Hence find $XY : YZ$.

Answer ..... : ..... [1]

Question 26 is printed on the next page

26



Box 1 Box 2

Box 1 contains 2 white balls. Box 2 contains 4 white balls and 3 black balls.

(a) Ann chooses, at random, one ball from each box.

(i) Find the probability that these balls are both black.

Answer ..... [1]

(ii) Find the probability that these balls have different colours.

Answer ..... [1]

(b) From the original contents of **Box 2**, Belle chooses, at random, two balls without replacement.

Find the probability that these balls are both white.

Answer ..... [1]

(c) Carla chooses one of the boxes at random.
With the original box contents, she then chooses, at random, one ball from this box.

Find the probability that the ball is white.

Answer ..... [2]

© UCLES 2016